

Human Interaction

“How do social environments and biological factors influence human behavior?”

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Data Selection/Exploration

Data Selection

- Measured behavioral attributes such as time spent alone, going outside, social acceptance, etc.
 - Measured psychological tendencies such as: feeling drained after socializing and stage fear.
 - We focused on the different tendencies of these behaviors between introverts and extroverts.
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- Rows: 2900
 - Columns: 8 variables

Data Selection/Exploration

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drained_after_socializing	Friends_circle_size	Post_frequency	Personality
0	4.0	No	4.0	6.0	No	13.0	5.0	Extrovert
1	9.0	Yes	0.0	0.0	Yes	0.0	3.0	Introvert
2	9.0	Yes	1.0	2.0	Yes	5.0	2.0	Introvert
3	0.0	No	6.0	7.0	No	14.0	8.0	Extrovert
4	3.0	No	9.0	4.0	No	8.0	5.0	Extrovert

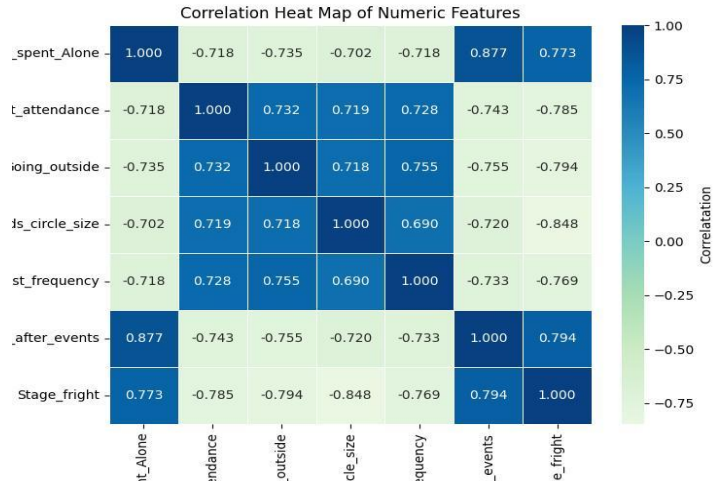
- 5 numerical values
 - 3 categorical values
 - 0 missing values
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- This analysis helps us clean and fully understand our data, especially because we have 0 missing values.

Data Wrangling

1. Checked for missing values within our data, output showed none.
2. Checked how many rows in our data were duplicated: 402 duplicates.
3. Created new variables so that our data analysis can be simplified: Drained_after_events and Stage_fright.

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drained_after_socializing	Friends_circle_size	Post_frequency	Personality	Drained_after_events	Stage_fright
0	4.0	No	4.0	6.0	No	13.0	5.0	Extrovert	1	1
1	9.0	Yes	0.0	0.0	Yes	0.0	3.0	Introvert	2	2
2	9.0	Yes	1.0	2.0	Yes	5.0	2.0	Introvert	2	2
3	0.0	No	6.0	7.0	No	14.0	8.0	Extrovert	1	1
4	3.0	No	9.0	4.0	No	8.0	5.0	Extrovert	1	1
...	...	...	...	...	...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No	6.0	6.0	Extrovert	1	1
2896	3.0	No	8.0	3.0	No	14.0	9.0	Extrovert	1	1
2897	4.0	Yes	1.0	1.0	Yes	4.0	0.0	Introvert	1	2
2898	11.0	Yes	1.0	3.0	Yes	2.0	0.0	Introvert	2	2
2899	3.0	No	6.0	6.0	No	6.0	9.0	Extrovert	1	1

Visual Insights



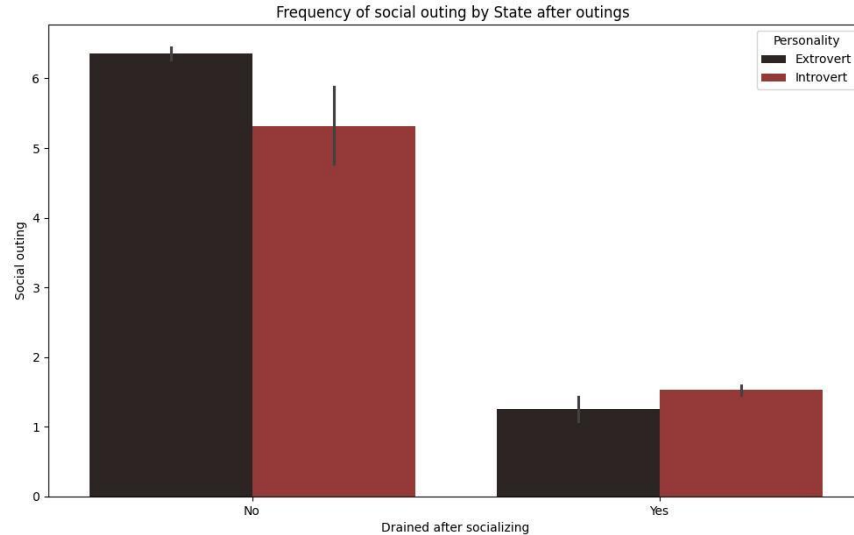
Strong negative correlations:

- Time spent alone and--->
- Going outside
- Social event attendance
- Friend circle size

Strongest Correlations:

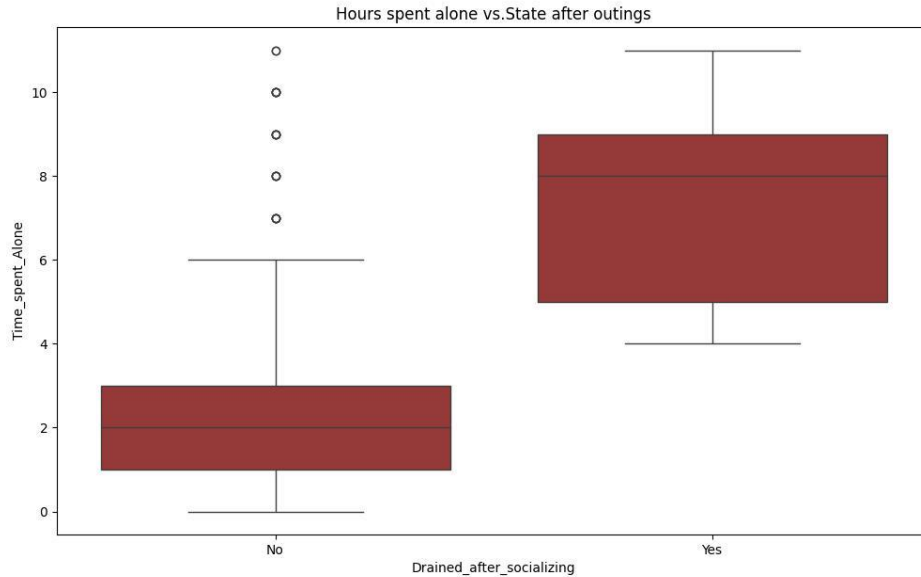
- Time_Spent_Alone and Drained_after_Events: strong positive correlation.
- Friend_circle_size and Stage_Fright: strong negative correlation

Bar Plot



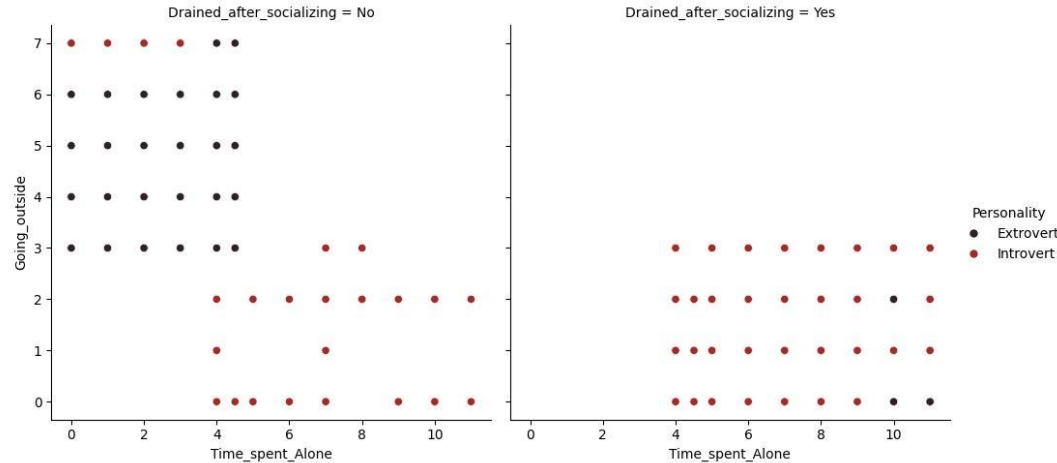
- Shows how much individuals engage in social outings after being socially drained.
- Regardless of the personality type, all individuals who feel more drained after socializing engage in social outings less.
- Barplot shows that sociability has a significant impact on social outings regardless of one's personality.

Boxplot



- Not drained=1-3 hrs of alone time
- Difference in solitary time required by different groups based on their energy levels
- Outliers shows unique cases where people do not feel drained but still choose to spend an unusually high amount of time alone.

Relational plot



- Negative trend
- color-coded points show that while Extroverts tend to be more active overall
- People who feel exhausted are clustered at the top-left, showing high solitude and low activity, while those who do not feel drained maintain a much higher spread of outdoor movement with minimal time alone

Conclusion

- Human interaction is not a random occurrence but a balanced system of energy and preference.
- Social environment + biological factors play significant roles in personality types.
- Because an individual's personality provides a biological baseline for how they process social stimulation, their state after outings acts as a gatekeeper for future movement.
- Those who feel biologically recharged by others are more likely to engage with the world, while those who feel drained after interactions require substantial solitary time to recover.
- These visualization prove that while environments provide the setting for interaction, our need for energy management is what dictates the rhythm of our social lives.